

Mantu Gupta

Curriculum Vitae

Decision Sciences Area
Indian Institute of Management Bangalore, India
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IIMB Profile

Education

- 2023–Present **Doctor of Philosophy (Ph.D.), Decision Sciences (Coursework CGPA: 3.52/4.00), *Indian Institute of Management Bangalore, India***, Advisor: Dr. Anand Deo.
Working on simulation and modeling of multivariate tail events using machine learning and Extreme Value Theory with application in estimating tail risk measures.
- 2018–2020 **M.Sc. in Applied Mathematics (CGPA: 8.30/10.00), *Indian Institute of Technology Mandi, India***, Advisor: Dr. Manoj Thakur.
Thesis: *Portfolio Optimization under Practical Constraints*. Applied classical mathematical methods and Multi-Objective Evolutionary Algorithms (NSGA-II, MOEA/D, MOGWO) to solve portfolio optimization problems with real-life constraints.
- 2015–2018 **B.Sc.(Honors) in Mathematics (CGPA: 8.16/10.00), *Sri Venkateswara College, University of Delhi, India***.

Research Interests

Multivariate Extreme Value Theory, Rare-Event Simulation, Machine Learning for Finance, Quantitative Risk Management, Portfolio Optimization

Working Papers

- 2025 **Self-Similar GEN: Simulation of Distribution Tails Using Deep Generative Models.**
With Dr. Anand Deo.
arXiv:2607.10700
- 2025 **Optimized Scheduling of Dual-Arm Single Cluster Tool for Concurrent Mixed-Wafer Processing in Semiconductor Manufacturing.**
With Dr. Reshma Chandrasekharan, Dr. Jitamitra Desai, and Amrutha Seshagiri.

Conference Presentations

- 2026 **Conference on Mathematical Finance (CMF-2026), IISER Pune, India.**
Self-Similar GEN: Simulating Tails of Heavy-Tailed Distributions Using Deep Generative Models. June 11–13, 2026.
- 2026 **International Workshop on Advances in Risk Analysis & Management: A Focus on Emergent Risks, Singapore.**
Self-Similar GEN: Simulating Tails of Heavy-Tailed Distributions Using Deep Generative Models. February 26–27, 2026.
- 2025 **INFORMS International Meeting, Singapore.**
Self-Similar Gen: Simulating Tails of Heavy-Tailed Distributions Using Generative Models. July 20–23, 2025.
- 2025 **India Management Research Conference (IMRC), Indian Institute of Management Ahmedabad.**
Self-Similar Gen: Simulating Tails of Heavy-Tailed Distributions Using Deep Generative Models. December 05–07, 2025.

Work Experience

- 2020–2023 **Researcher**, *TCS Research & Innovation*, Hyderabad, India.
Developed machine learning and computer vision algorithms for structured information extraction. End-to-end development of PoCs and deployed applications. Filed multiple patents and received On-the-Spot, IP Creation, and Special Initiative Awards.

Patents

- 2021 **Extracting Region of Interest from Scanned Images and Determining Image Type**. Co-Inventor. Granted in the US and EPO; Pending in India.
- 2022 **Method and system for detecting and extracting price region from digital flyers and promotions**. Co-Inventor. Granted in the US; Pending in India and EPO.
- 2023 **Method and system for identifying product details from marketing collateral images**. Co-Inventor. Pending in the US, EPO, and India.

Technical Skills

Programming	Python, Java, R, MATLAB
ML/DL	PyTorch, TensorFlow, Scikit-learn
Optimization	Gurobi, Mixed-Integer Programming
Tools	Linux, Git, $\text{\LaTeX}$

Teaching

- 2026 Python for Data Science (PGPBA Preparatory), Indian Institute of Management Bangalore. May 2026.

Teaching Assistance

- 2025–26 Decision Sciences I (PGP), Jun–Aug.
- 2025–26 Quantitative Risk Management (PGP), Dec–Apr.
- 2026–27 Optimization (PGPBA), June–Aug.

Workshops & Training

- 2025 **ACM India Winter School 2025: AI and Finance**, International Institute of Information Technology Hyderabad.
- 2025 **Large Scale Optimization Summer School 2025**, IEOR, Indian Institute of Technology Bombay.
- 2025 **Workshop on Advances in Data-Driven Optimization**, Brij Disa Centre for Data Science and AI, Indian Institute of Management Ahmedabad.
- 2025 **exSPLORe 2025 — Explorations in Statistics, Probability, Learning and Optimisation Research**, CDLDS, Ashoka University, Sonipat.

Scholarships & Fellowships

- 2023–Present Doctoral Fellowship, Indian Institute of Management Bangalore.

Competitive Examinations

- 2024 GATE 2024 — Data Science and Artificial Intelligence (DA). Qualified.
- 2023 GATE 2023 — Mathematics (MA). Qualified.
- 2022 GATE 2022 — Mathematics (MA). Score: 656, AIR 105 (top 1%).

- 2018 IIT JAM 2018 — Mathematics. Qualified.
2018 Delhi University Entrance Test (M.Sc.). AIR 18 (top 1%).

Awards & Honors

- 2023 IP Creation Awards — TCS Research.
2022 Special Initiative Award — TCS Research.
2021 On-the-Spot Award — TCS Research.
2013 Certificate of Merit for AISSE (CBSE).

Academic Service

- 2024–25 PhD Student Academic Representative, IIM Bangalore. Designed and managed academic timetables using Mixed-Integer Linear Programming (MILP).
2024–25 Student Coordinator, Brown Bag Seminar Series, IIM Bangalore.

Languages

English (Professional Working), Hindi (Full Professional), Odia (Limited Working), Bhojpuri (Native)

References

Dr. Anand Deo

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Dr. Manoj Thakur

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